Quaternion orientation time derivative derivation

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Report

Can you walk me through the derivation for the time derivative of a Quaternion orientation? Representing the quaternion orientation with an axis angle form

Time Derivative of a Quaternion Orientation

Setup: Quaternion from Axis-Angle

A unit quaternion representing a rotation of angle θ about unit axis **n̂** = (n_x, n_y, n_z) is:

$$q = \cos\frac{\theta}{2} + \sin\frac{\theta}{2}(n_x\mathbf{i} + n_y\mathbf{j} + n_z\mathbf{k})$$

Or compactly:

$$q = \begin{bmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2}\,\hat{n} \end{bmatrix} = \begin{bmatrix} q_w \\ \mathbf{q}_v \end{bmatrix}$$

Step 1: Introduce Angular Velocity

The body's angular velocity vector is defined as:

$$\boldsymbol{\omega} = \dot{\theta}\,\hat{n}$$

(rate of change of angle times the axis — this assumes the axis is momentarily fixed, which is valid instantaneously.)

Step 2: Differentiate the Quaternion Directly

Taking $\dot{q}$, we need to differentiate both components. First, consider an infinitesimal rotation over time dt:

$$q(t + dt) = \delta q \otimes q(t)$$

where δq is the quaternion for an infinitesimal rotation of angle $\omega\, dt$ about $\hat{\boldsymbol{\omega}}$:

$$\delta q = \begin{bmatrix} \cos \frac{\omega\, dt}{2} \\ \sin \frac{\omega\, dt}{2}\, \hat{\omega} \end{bmatrix} \approx \begin{bmatrix} 1 \\ \frac{1}{2}\boldsymbol{\omega}\, dt \end{bmatrix}$$

using small-angle approximation ($\cos \epsilon \approx 1, \sin \epsilon \approx \epsilon$).

Step 3: Expand the Product

$$q(t + dt) - q(t) = (\delta q - \mathbf{1}) \otimes q(t)$$

$$\dot{q} = \lim_{dt \to 0} \frac{q(t + dt) - q(t)}{dt} = \frac{1}{2} \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix} \otimes q(t)$$

Writing Ω as the "pure quaternion" formed from angular velocity:

$$\Omega = \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix}$$

$$\boxed{\dot{q} = \frac{1}{2}\, \Omega \otimes q}$$

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